

Thiranex Data Analytics Internship Task 3

Predictive Analysis using Sales Data

By Syeda Sidrah Amena

Introduction

This project focuses on analyzing historical retail sales data and forecasting future sales trends using predictive analytics techniques. The objective was to identify sales patterns, generate future predictions, and visualize business insights through an interactive Power BI dashboard.

Project Workflow

The dataset was first cleaned and preprocessed by handling missing values, converting date formats, and organizing sales data into monthly trends. Historical sales patterns were then analyzed to identify fluctuations and overall growth trends.

A Linear Regression model was used to forecast future sales based on historical monthly sales data. Model performance was evaluated using standard accuracy metrics such as MAE, RMSE, and R^2 Score.

The results were visualized using Power BI through KPI cards, sales trend charts, forecast visualizations, and category-wise sales analysis.

Dashboard Highlights

- Historical monthly sales analysis
- Forecasted sales trend visualization
- KPI cards for sales insights
- Category-wise sales comparison
- Interactive slicers for filtering data

Conclusion

The project successfully demonstrated how predictive analytics can be used to forecast future sales trends using historical data. The insights generated through forecasting and visualization can support data-driven business planning and decision-making.